

The Morning After

Assessing the impact of a major default

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POLICY RESPONSE TO THE FINANCIAL CRISIS

OTC DERIVATIVES AGENDA BEFORE 2008

Backlogs of unconfirmed trades

Increased transparency regarding counterparties and exposures

Creation of trade repository

THE POST-CRISIS PROGRAM

Clearing of derivatives at CCPs

Collateralization of uncleared derivatives

Revamped prudential regulation

Restrictions on collateral re-use

BENEFITS

Balance
sheet
improvement

Risk
reduction

Trading
Benefits

Operational
benefits

-VS-

C
C
P

COSTS

Risk
concentration

Systemic
risk

Costs

Collateral

The Morning After THE CCP WORLD

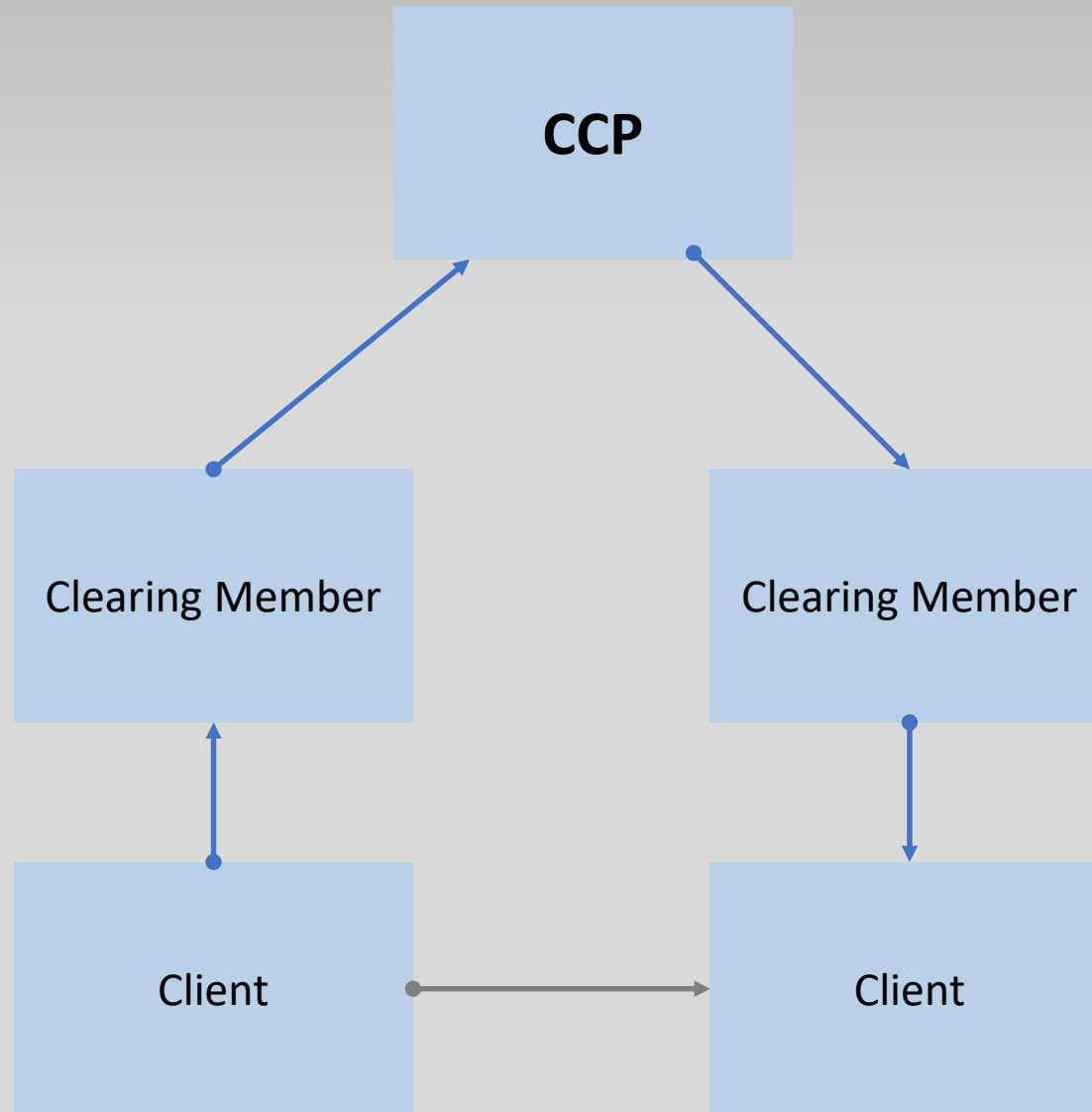
FROM 3 CCPs AND 1 TR IN 2008...



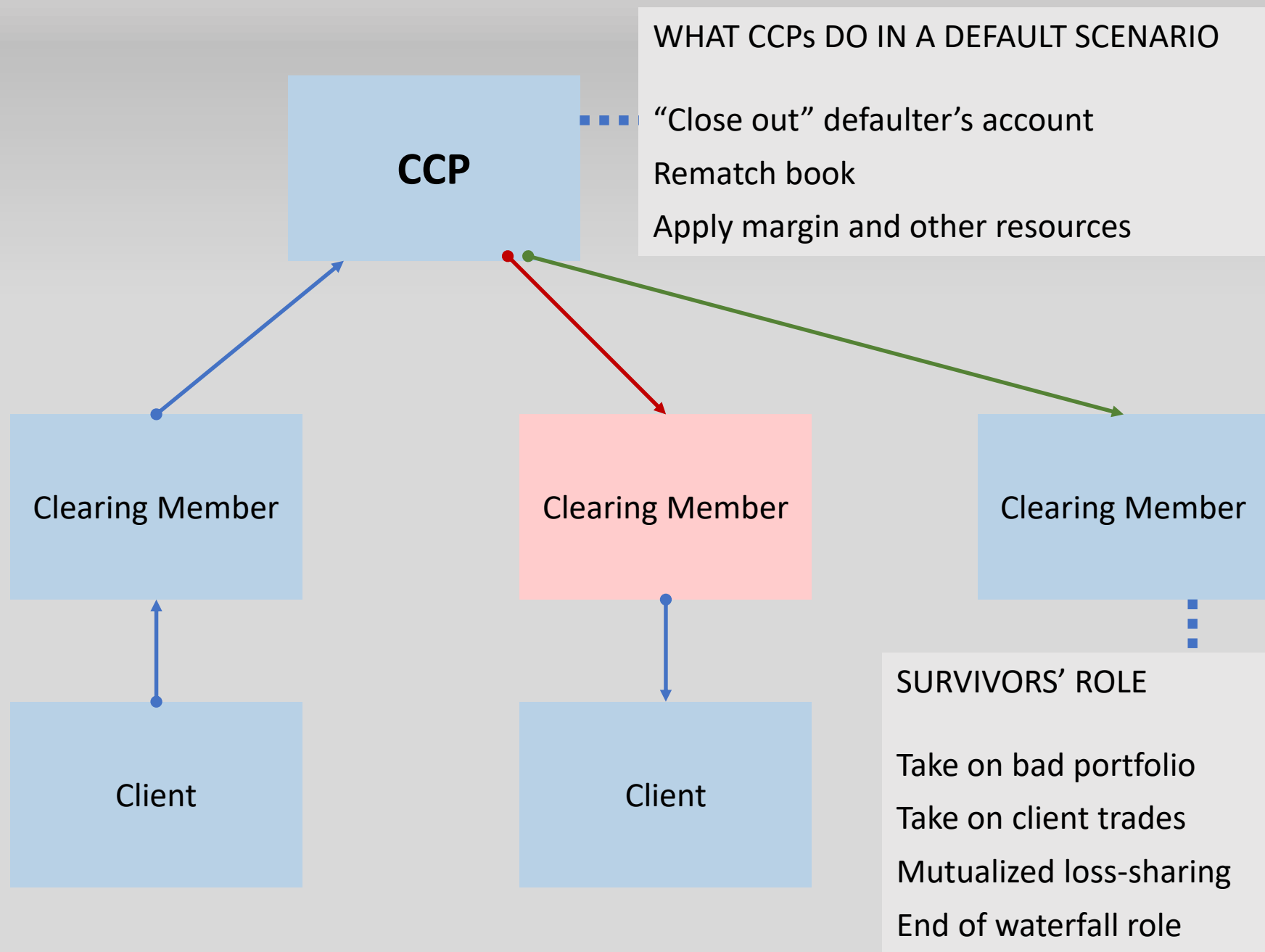
Source: FSB, OTC Derivatives Market Reforms, 6th progress report on implementation, September 2013

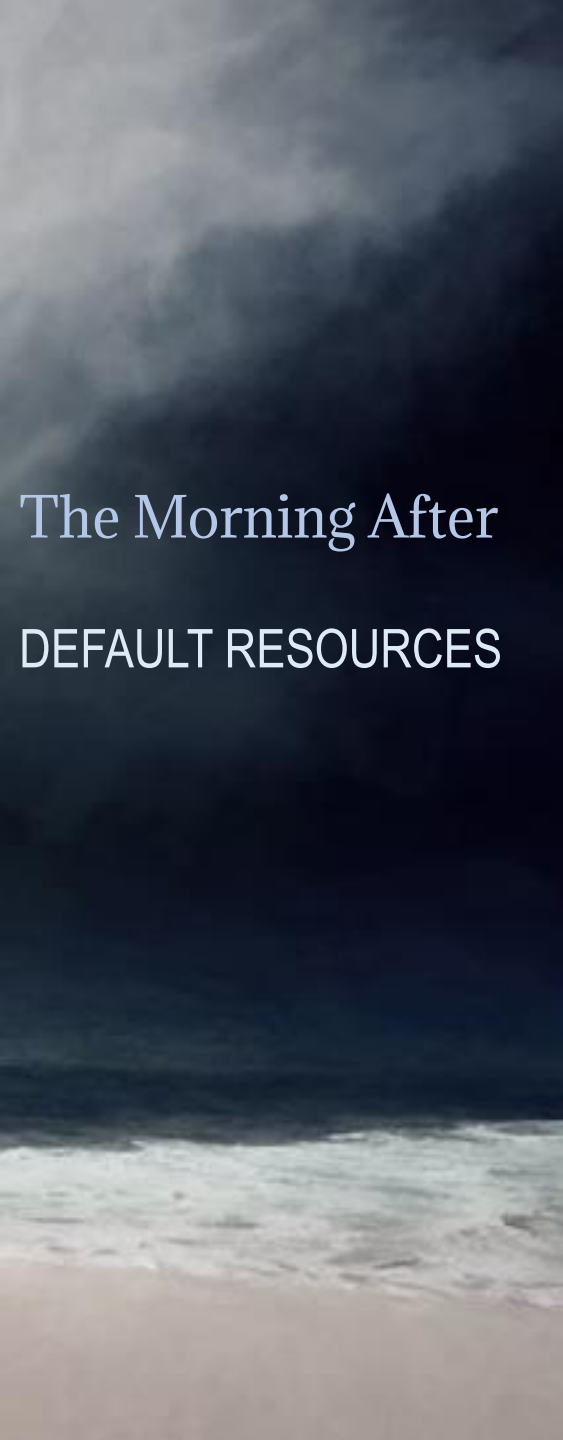
● CCPs
● TRs

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DERIVATIVES CLEARING



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DERIVATIVES CLEARING
- DEFAULT





DEFAULT RESOURCES

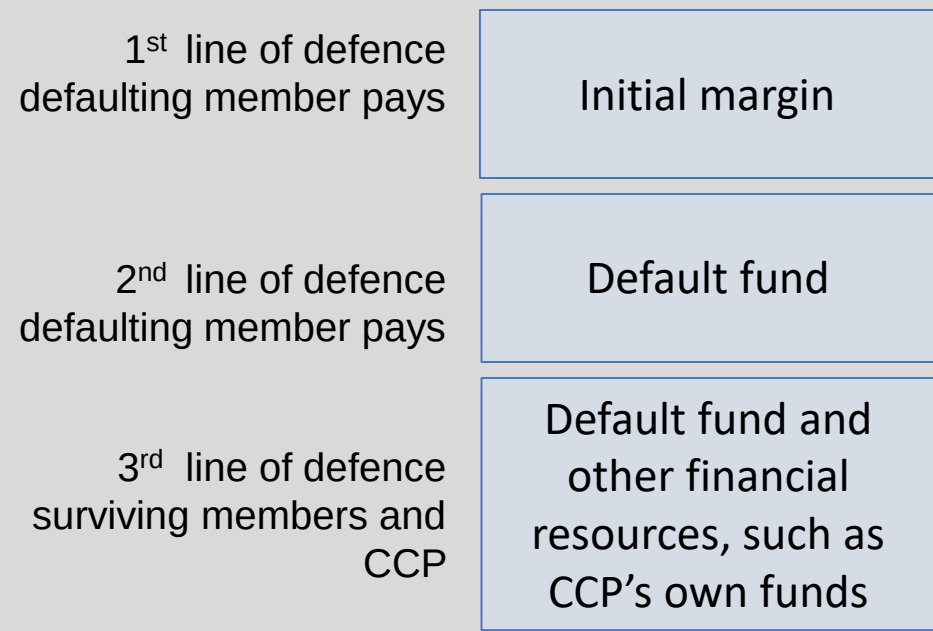
Prefunded

Waterfall

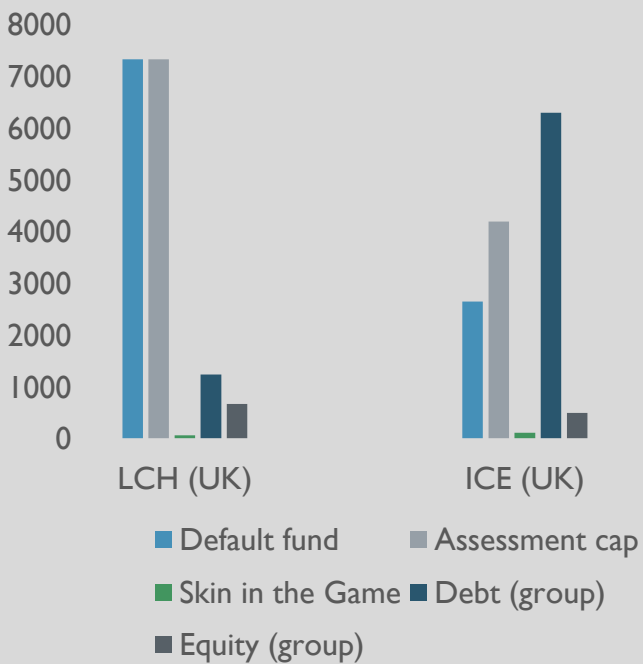
End of waterfall

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DEFAULT RESOURCES



Waterfall resources (USD, millions)



The Morning After OVERFLOW RISK

OVERFLOWING THE WATERFALL

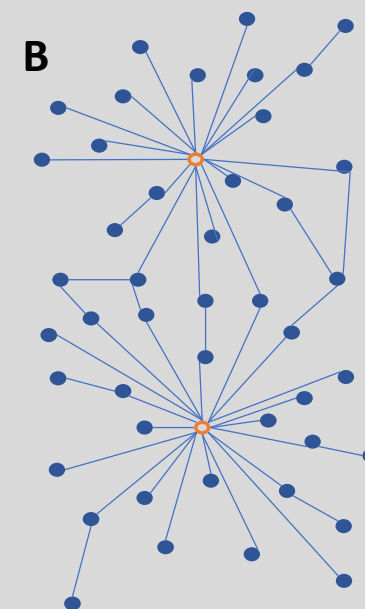
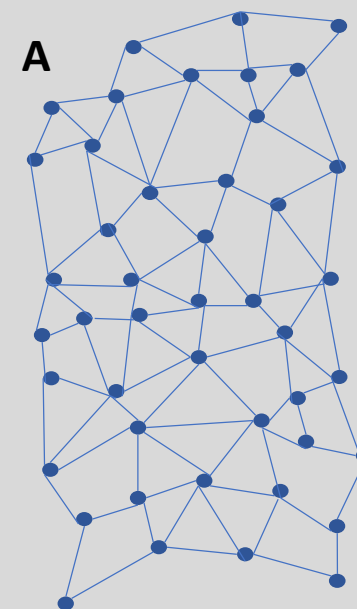
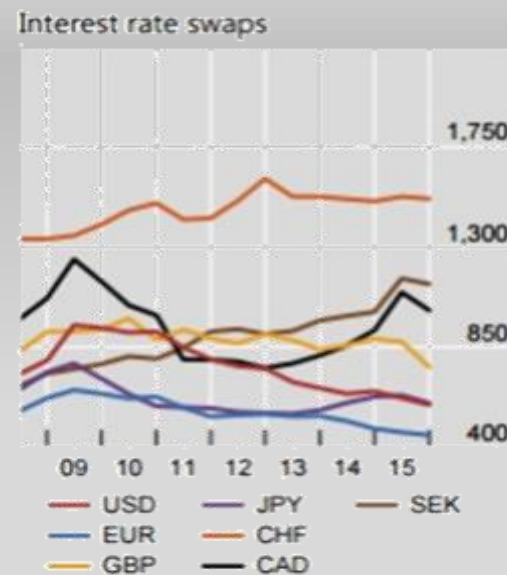
Margin is not intended to contain all defaults

Post-crisis instances of clearing member default have utilized default fund

Cover-2 standard has been shown to be insufficient

The post-crisis market is more concentrated and less resilient

Mandatory clearing still not fully rolled-out





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RECOVERY AND RESOLUTION

WHEN THE WATERFALL IS GONE

Pass round the hat

Variation Margin Gains Haircutting

Tear up

BANK RECOVERY AND RESOLUTION TOOLS

■ ■ ■ ■ Bail-in of wholesale debt

Business transfer and 'bridge bank' (good bank/bad bank)

THESE DON'T WORK FOR CCPs

RECOVERY AND RESOLUTION - POLICY

"Restoring the ability of the CCP to perform its critical functions as a going concern"

... but ...

"Continuity of critical functions" implies CCP should not be allowed to fail: where clearing is mandatory, "criticality" is axiomatic ?

■ ■ ■ ■ AVOIDING TAXPAYER BAILOUT

Keep the losses within the CM/User community = Orderly Loss Distribution

VMGH can contain loss, if financially uncapped (RBA research)



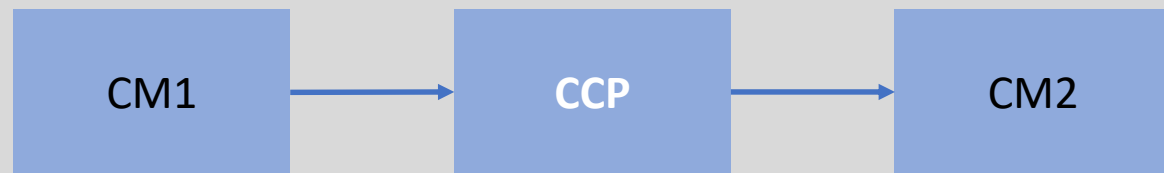
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VMGH

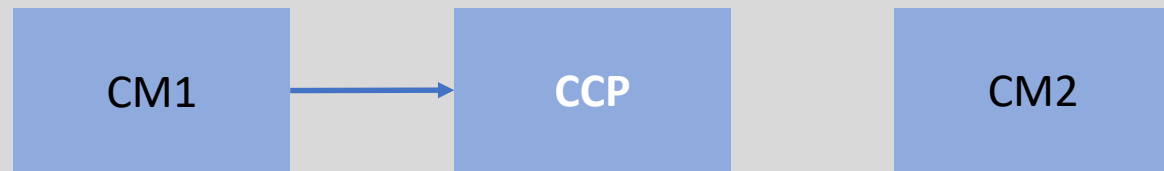
VARIATION MARGIN GAINS HAIRCUTTING

VM = Daily catch-up payment to reflect to-date change in value

VMGH = Non-payment to gainers



Normal conditions



VMGH

Losers still pay
Gainers forego gains



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THE STORY SO FAR

TEN YEARS AFTER LEHMANS – A STOCKTAKE

Clearing and collateralization of standard derivatives is the norm

Bank recovery and resolution procedures have been overhauled; taxpayer bailouts, even of infrastructures, are ruled out

Prudential regulation (and investor protection regulation) have been overhauled – new prudential measures introduced

The post-trade market is highly concentrated and becoming more so

Demand for HQLA (collateral) has been rising: margin rules, cash liquidity, liquidity coverage rules

... IMPLICATIONS FOR HANDLING THE NEXT MAJOR DEFAULT ?



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PRUDENTIAL REGULATION AND COLLATERAL

SELECTED REGULATORY REFORMS

Clearing of standard derivatives

- Initial margin flows upward only

Collateralization of uncleared derivatives

- Gross transfers
- No re-use allowed

Liquidity coverage ratio

- Fund of HQLA to insure against peak outflows of cash

IMF WP 18/65: CCP Resolution

IMF WP 18/228: Morning after

PRUDENTIAL MEASURES AFFECTED BY A MAJOR DEFAULT

Capital deduction – losses

Counterparty risk capital requirement

Position risk capital requirement

Credit Valuation Adjustment

Operational risk capital requirement

Large exposure requirement

Leverage ratio

Liquidity coverage ratio



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SUMMARY OF IMF
18/228

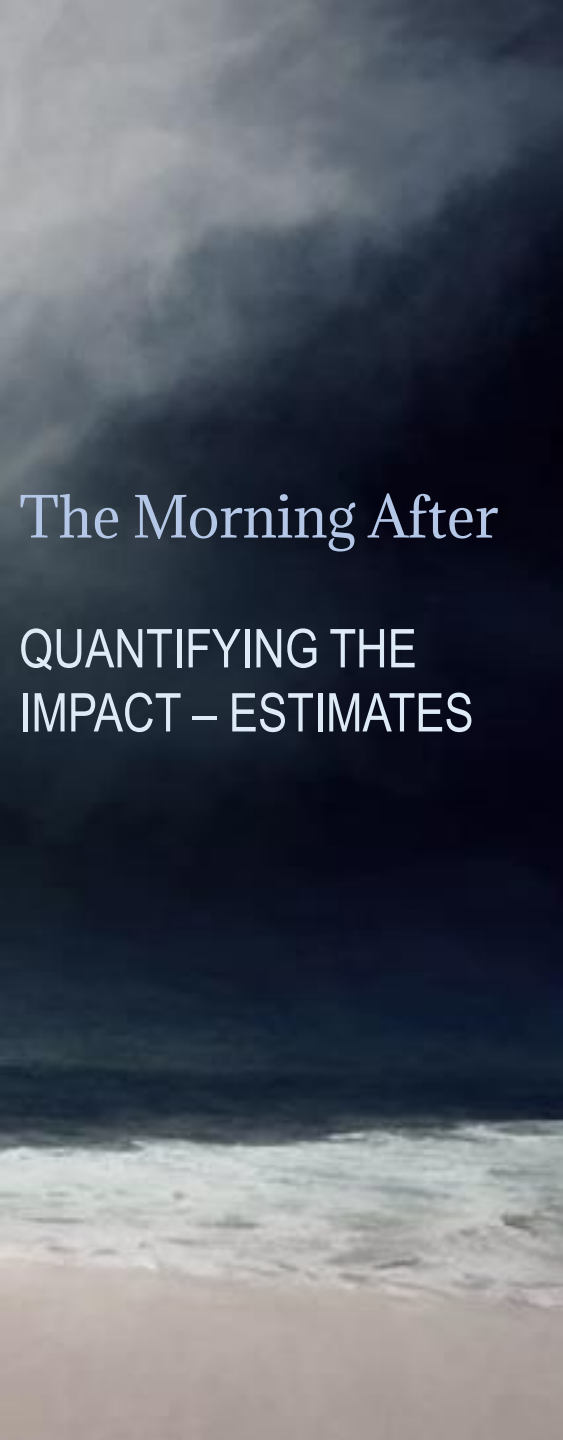
Factors including prudential metrics (e.g., liquidity coverage ratio) obliges firms to set aside HQLA (high quality liquid assets)

Immediately following a major default, the amount of HQLA demanded by the whole market would spike

Actions by CCPs following a default, which would pass through to clearing members and their clients, could be managed. VMGH would appear still to be the least unpleasant remedy for a CCP in difficulty

We estimate the size of the spike and draw conclusions as to whether the depth of the market is adequate to absorb it

Estimates no more than \$100 billion of HQLA following a major default



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QUANTIFYING THE IMPACT – ESTIMATES

(U.S. dollars in billions)	Direct Demand for HQLA	LCR-Related Demand for HQLA	Total
Default fund replenishment (multi-CCP)	-	22	22
Initial margin increases (multi- CCP)	25	-	25
Initial margin increases (uncleared market)	13	0.7	13.7
Variation margin foregone (closed-out but unvalued bilateral trades with defaulter)	-	12	12
VMGH (single CCP)	-	18.5	18.5
Total		53.2	91.2



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CONCLUSIONS

Around \$ 90 bn additional demand for HQLA strikes the market within the first few days

- Of this, about 60% is attributable to LCR rules
- In context, top-12 banks have stocks of HQLA ~ \$ 350 bn each
- Total available supply of HQLA > \$ 6,000 bn, of which around \$ 1,500 bn held by banks

It's a shock – no surprise – but containable

Limited VMGH is validated as end-of-waterfall strategy for CCPs

Monetary and procyclicality implications of prudential policy deserve further analysis